SCETATHON 2026

Team Id - SCET2026-029

Team Name - Array of Sunshine

Project Title – CampusXR: Hierarchical 360° Navigation with Experimental 3DGS Mode

Problem Statement Title - Virtual Campus Explorer

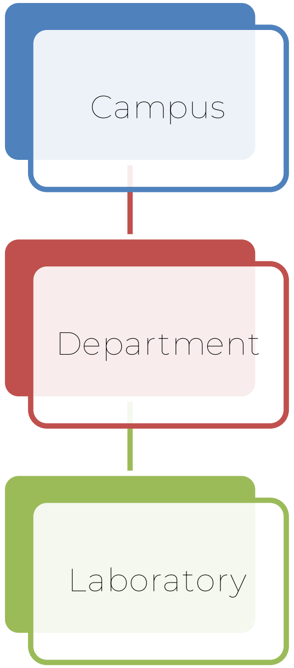
SCET2026-029

CampusXR

Hierarchical 360° Navigation with Experimental 3DGS Mode

Proposed Solution

- Web-based immersive virtual campus platform
- Structured navigation: **Campus → Department → Laboratory**
- High-resolution 360° panoramas with interactive hotspots
- Experimental **3D Gaussian Splatting (3DGS)** mode
- Admin-managed content system (no coding required)



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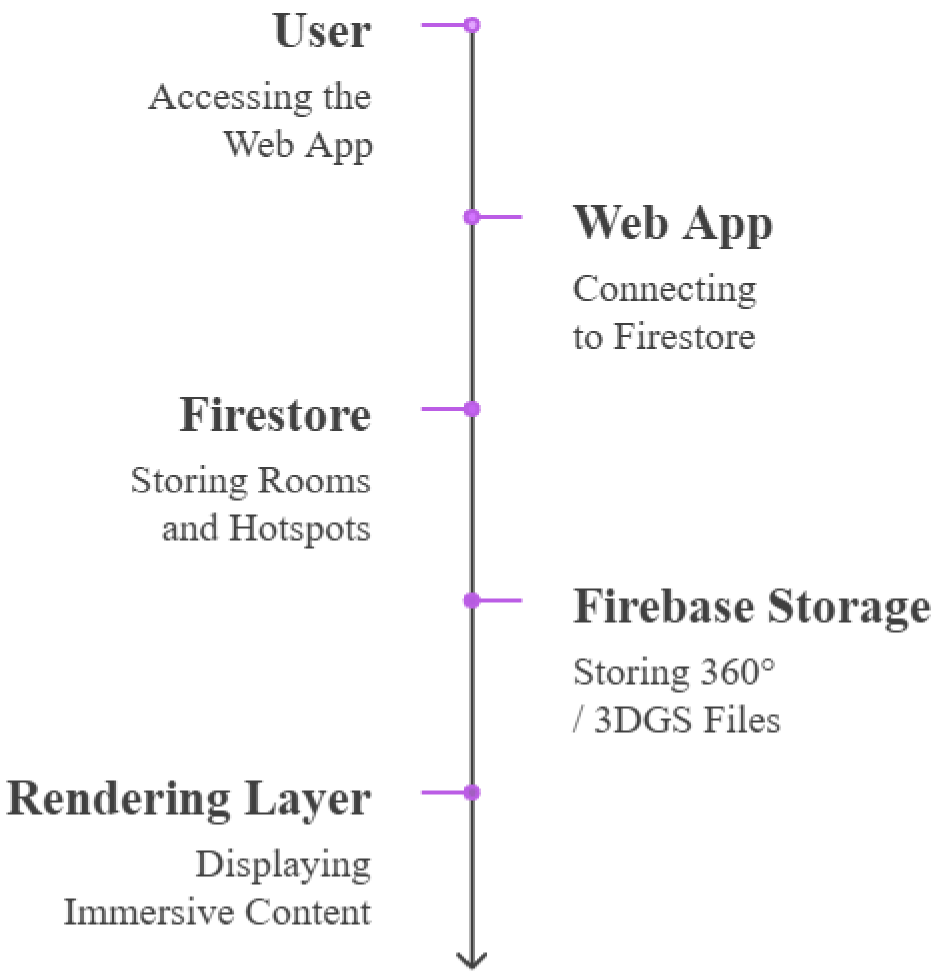
TECHNICAL APPROACH

Technologies Used

- Frontend: React (Vite)
- Styling: Tailwind CSS v4
- Animations: GSAP + @gsap/react
- 360° Viewer: Pannellum (CDN)
- 3DGS Rendering: Three.js
- Backend: Firebase (Firestore, Storage, Auth)
- Hosting: Firebase Hosting

System Architecture Flow

- User selects Room
- Web App fetches room data + hotspots from Firestore
- Panorama loads via Pannellum
- Optional 3DGS mode rendered via Three.js
- Admin Panel writes updates directly to Firestore



SCET2026-029

FEASIBILITY AND VIABILITY

Feasibility Analysis

- 24-hour optimized development roadmap
- Pre-prepared 360° and 3DGS demo assets
- Firebase free-tier deployment (no backend setup)
- Desktop-first scope reduces complexity
- Limited demo rooms (≤10) for stable performance

Potential Challenges

- Large 3DGS file loading time
- Firebase free-tier storage & read quota limits
- Hotspot coordinate accuracy
- Browser compatibility issues

Mitigation Strategies

- Compress and optimize 360° images
- Preload / cache 3DGS scenes before demo
- Early hotspot testing in Admin Panel
- Chrome as primary demo browser

SCET2026-029

IMPACT AND BENEFITS

Impact on Target Audience

- Enables immersive pre-visit campus exploration from anywhere
- Increases decision confidence for prospective applicants
- Improves accessibility for users with mobility constraints
- Enhances institutional digital presence and global outreach

Key Benefits

Social

- Inclusive digital campus access
- Reduces mobility and geographic barriers

Economic

- Reduces campus visit costs
- Enhances institutional marketing reach

Technological

- Showcases next-generation neural rendering (3DGS)
- Bridges traditional virtual tours with Neural 3D reconstruction (Gaussian Splatting)

SCET2026-029

RESEARCH AND REFERENCES

Technical Documentation

- Firebase Documentation (Firestore, Storage, Auth, Hosting)
- Pannellum (MIT Licensed 360° Panorama Viewer)
- Three.js Official Documentation
- GSAP Animation Library

Research & Innovation Basis

- 3D Gaussian Splatting - Neural Rendering Research
- Real-time Web-based 3D Rendering Techniques
- Hierarchical Navigation UX Design Principles